

## HUGHENDEN AGCM, AB, Canada

WMO: 717940

Lat: 52.5781N

Lon: 110.7836W

Elev: 703

StdP: 93.16

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 12            | -31.0      | -28.2 | -34.3                         | 0.2 | -30.8 | -31.3 | 0.2 | -28.1 | N/A                   | N/A  | N/A | N/A  | N/A                   | N/A  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.3                   | 28.7            | 16.9 | 26.7 | 16.2 | 25.2 | 15.7 | 18.9                | 25.3 | 17.8 | 24.2 | 16.8 | 22.9 | N/A                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.5                            | 12.8 | 21.9 | 15.4 | 11.9 | 20.6 | 14.5 | 11.2 | 19.4 | 56.8          | 25.2 | 53.1 | 24.5 | 49.9 | 23.1 | 23.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| N/A               | N/A  | N/A | -34.6                      | 32.4 | 1.8                | 2.0 | -35.9                                              | 33.9 | -36.9      | 35.1 | -37.9      | 36.2 | -39.2      | 37.7 |  |  |
|                   |      |     | -34.7                      | 21.1 | 1.7                | 1.5 | -36.0                                              | 22.2 | -37.0      | 23.1 | -38.0      | 24.0 | -39.3      | 25.1 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 2.9   | -11.0 | -12.0 | -4.9  | 3.2   | 10.8  | 14.9  | 17.1  | 16.4  | 11.7  | 4.1   | -4.8  | -12.0 | (6)  |
| (7)  |                                                                                  | DBStd       | 12.37 | 8.27  | 7.76  | 7.49  | 5.25  | 4.39  | 3.08  | 2.68  | 3.19  | 4.52  | 5.03  | 7.10  | 7.88  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 3333  | 651   | 615   | 462   | 211   | 43    | 3     | 0     | 1     | 33    | 191   | 443   | 681   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 5682  | 909   | 848   | 720   | 454   | 233   | 108   | 56    | 76    | 204   | 441   | 693   | 939   | (9)  |
| (10) |                                                                                  | CDD10.0     | 738   | 0     | 0     | 0     | 6     | 69    | 151   | 221   | 199   | 83    | 9     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD18.3     | 45    | 0     | 0     | 0     | 0     | 1     | 7     | 18    | 15    | 4     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 818   | 0     | 0     | 0     | 3     | 65    | 105   | 269   | 253   | 121   | 3     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 177   | 0     | 0     | 0     | 0     | 7     | 18    | 61    | 53    | 37    | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 385   | 19    | 11    | 15    | 28    | 40    | 70    | 71    | 52    | 29    | 17    | 15    | 15    | (15) |
| (16) |                                                                                  | PrecMax     | 522   | 39    | 23    | 35    | 81    | 86    | 120   | 137   | 127   | 73    | 41    | 50    | 28    | (16) |
| (17) |                                                                                  | PrecMin     | 242   | 5     | 4     | 1     | 2     | 6     | 23    | 25    | 6     | 0     | 2     | 3     | 4     | (17) |
| (18) |                                                                                  | PrecStd     | 77    | 9     | 5     | 8     | 18    | 24    | 27    | 26    | 26    | 22    | 9     | 12    | 6     | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 5.8   | 5.5   | 13.7  | 22.0  | 27.8  | 29.1  | 30.7  | 30.4  | 30.7  | 22.9  | 13.6  | 5.3   | (19) |
| (20) |                                                                                  |             | MCWB  | 2.4   | 2.4   | 6.6   | 9.6   | 13.4  | 17.2  | 19.5  | 16.9  | 15.3  | 12.7  | 6.9   | 2.0   | (20) |
| (21) |                                                                                  | 2%          | DB    | 4.1   | 3.0   | 10.1  | 18.4  | 25.3  | 26.0  | 28.3  | 28.2  | 26.7  | 17.9  | 9.0   | 3.3   | (21) |
| (22) |                                                                                  |             | MCWB  | 1.6   | 0.9   | 4.6   | 8.4   | 12.7  | 15.6  | 18.2  | 17.3  | 14.3  | 10.4  | 4.4   | 0.9   | (22) |
| (23) |                                                                                  | 5%          | DB    | 2.5   | 1.3   | 6.7   | 15.2  | 22.7  | 24.1  | 26.3  | 26.2  | 23.7  | 14.9  | 6.0   | 0.8   | (23) |
| (24) |                                                                                  |             | MCWB  | 0.4   | -0.4  | 3.1   | 6.7   | 11.8  | 15.3  | 17.3  | 16.6  | 13.6  | 8.5   | 2.9   | -1.2  | (24) |
| (25) |                                                                                  | 10%         | DB    | 0.6   | -0.6  | 4.1   | 12.4  | 20.1  | 22.2  | 24.4  | 24.2  | 20.6  | 12.0  | 3.7   | -1.5  | (25) |
| (26) |                                                                                  |             | MCWB  | -1.0  | -1.9  | 1.4   | 5.4   | 10.5  | 14.2  | 16.7  | 16.2  | 12.3  | 6.7   | 1.4   | -2.8  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 2.8   | 2.6   | 7.5   | 10.5  | 14.8  | 18.7  | 21.3  | 20.0  | 16.7  | 13.2  | 7.4   | 2.6   | (27) |
| (28) |                                                                                  |             | MCDWB | 5.3   | 4.9   | 13.1  | 21.4  | 25.2  | 25.6  | 27.9  | 26.2  | 25.6  | 21.8  | 12.4  | 4.9   | (28) |
| (29) |                                                                                  | 2%          | WB    | 1.7   | 1.0   | 4.9   | 8.6   | 13.4  | 17.1  | 19.3  | 18.7  | 15.2  | 11.1  | 4.8   | 1.0   | (29) |
| (30) |                                                                                  |             | MCDWB | 3.8   | 2.7   | 9.3   | 17.7  | 22.5  | 23.9  | 25.7  | 25.5  | 24.6  | 16.2  | 8.6   | 3.3   | (30) |
| (31) |                                                                                  | 5%          | WB    | 0.6   | -0.3  | 3.2   | 7.1   | 12.4  | 15.9  | 18.2  | 17.5  | 14.0  | 9.0   | 3.0   | -1.0  | (31) |
| (32) |                                                                                  |             | MCDWB | 2.5   | 1.2   | 6.5   | 14.1  | 21.1  | 21.9  | 24.4  | 23.8  | 22.4  | 14.5  | 5.5   | 0.8   | (32) |
| (33) |                                                                                  | 10%         | WB    | -1.1  | -2.0  | 1.6   | 5.7   | 11.3  | 14.9  | 17.3  | 16.5  | 12.9  | 7.4   | 1.5   | -2.8  | (33) |
| (34) |                                                                                  |             | MCDWB | 0.5   | -0.7  | 4.0   | 11.6  | 19.2  | 20.7  | 23.2  | 22.6  | 20.1  | 11.8  | 3.5   | -1.4  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 9.7   | 9.9   | 9.8   | 11.4  | 13.1  | 11.5  | 12.3  | 12.8  | 13.4  | 11.2  | 9.0   | 8.6   | (35) |
| (36) |                                                                                  |             | MCDBR | 10.3  | 10.8  | 12.6  | 16.8  | 17.0  | 14.6  | 16.0  | 16.5  | 19.2  | 16.2  | 12.0  | 10.6  | (36) |
| (37) |                                                                                  |             | MCWBR | 8.3   | 9.0   | 8.1   | 8.4   | 7.5   | 6.3   | 7.4   | 7.3   | 8.5   | 9.0   | 8.0   | 8.6   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 10.5  | 10.7  | 12.2  | 15.6  | 15.1  | 12.4  | 14.0  | 14.2  | 17.3  | 15.2  | 11.0  | 10.2  | (38) |
| (39) |                                                                                  |             | MCWBR | 8.6   | 9.0   | 8.1   | 8.1   | 7.2   | 6.0   | 7.1   | 6.9   | 8.1   | 8.7   | 7.6   | 8.5   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.250 | 0.296 | 0.287 | 0.333 | 0.356 | 0.372 | 0.362 | 0.376 | 0.317 | 0.281 | 0.257 | 0.238 | (40)  |      |
| (41) |                                                                                  | taud        | 2.344 | 2.236 | 2.436 | 2.353 | 2.341 | 2.330 | 2.394 | 2.329 | 2.465 | 2.544 | 2.460 | 2.354 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 768   | 811   | 916   | 902   | 894   | 880   | 882   | 847   | 868   | 843   | 763   | 720   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 57    | 88    | 88    | 109   | 117   | 120   | 110   | 111   | 86    | 64    | 51    | 47    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.09  | 2.15  | 3.61  | 4.75  | 5.79  | 5.97  | 6.31  | 5.12  | 3.62  | 2.16  | 1.09  | 0.82  | (44)  |      |
| (45) |                                                                                  | RadStd      | 0.10  | 0.16  | 0.27  | 0.45  | 0.39  | 0.41  | 0.31  | 0.42  | 0.38  | 0.29  | 0.11  | 0.07  | (45)  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (62 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page